What Personalised Reward Models Learn About a New Rater: Scale Use, and a Detection Limit for Content Preferences

Anonymous Authors[1]

Abstract

Personalised reward models are meant to learn how individual people judge model outputs. We test what they learn about raters they have never seen, with one protocol on three multi-rater datasets (MultiPref, DICES, PRISM): k of a new rater's ratings for adaptation, held-out items for evaluation, and a permutation control that compares a rater's own adapted parameters with another rater's. Across group-conditioned, low-rank, published (VPL, PAL, LoRe), in-context and fine-tuned personalisation, what transfers to a new rater is scale use: a text-independent baseline with two identifiable offsets per rater gains 0.08–0.10 nats per rating at k = 20 on all three datasets, is matched only by a tuned low-rank latent, and loses its gain on binary labels, as do all methods (at most 0.002 nats). No method improves which response a new rater prefers: on MultiPref and on 41,199 native PRISM choices, every method's gain over its unpersonalised version has an upper 95 % bound below one point. A semi-synthetic positive control locates the reason. These methods detect injected content preferences only when an oracle would gain about +10 (full features) to +21 (compressed features) points of direction accuracy, whereas a learner told the right content features recovers about 60 % of a +5-point preference from the same 20 ratings: the bottleneck is finding the content dimension, not the number of ratings. We recommend graded labels, new-rater evaluation with a permutation control, separate calibration and ranking metrics, and a scale-use baseline.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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1. Introduction

Reward models and LLM judges are trained on human ratings that disagree. A recent line of work treats this disagreement as signal to be modelled rather than noise to be averaged: distributional preference learning estimates a distribution over each alternative's utility rather than a point score (Siththaranjan et al., 2024); jury learning predicts individual annotators from content, annotator and demographic-group embeddings (Gordon et al., 2022), and large datasets such as PRISM are analysed by sociodemographic group (Kirk et al., 2024); personalised reward models learn a per-user latent, inferred for a new user from a few examples (Poddar et al., 2024; Chen et al., 2025). The pluralistic-alignment agenda (Sorensen et al., 2024) makes the representation of individual and group differences a first-class goal.

What these models learn about a person is rarely tested directly. Evaluations typically use synthetic users whose preferences differ by construction, binary preference labels, or users seen in training. We ask a simple question under a common protocol on real raters: **given k ratings from someone the model has never seen, how much better can it predict their other ratings — in particular, which response they prefer — and what is it that it learns?** The answer we find is that the methods learn calibration, not ranking — and a semi-synthetic positive control shows how strong a rater's content preferences would have to be before they learn ranking.

Measurement theory suggests a candidate answer that reward modelling largely ignores: raters differ in **how they use the response scale** — some say "clearly better" where others say "slightly better", some rarely call a tie, some are harsher across the board (Linacre, 1989; Baumgartner & Steenkamp, 2001). Such response styles are text-independent, cheap to estimate, and invisible to models that condition only on content or demographics. We make them a reference point for calibration. Because they are text-independent they barely change which response a rater is predicted to prefer (only through asymmetric "clearly"/"slightly" mass); a personalised reward model that learns content preferences should therefore show its value in ranking, and a model whose likelihood gain does not exceed the scale-use offsets has not shown that it learns more than

scale use.

Contributions. (1) A new-rater evaluation protocol for personalised reward models — held-out raters and items, k-shot adaptation, a permutation control (own versus another rater's adapted parameters) — and a scale-use baseline (a few text-independent offsets per rater; two identifiable on two-sided preferences) that plugs into any graded reward model (§4–5). (2) A systematic test on three datasets of group-conditioned, low-rank, published (VPL, PAL, LoRe), in-context and fine-tuned personalisation: what transfers is scale use, which is calibration rather than ranking and disappears on binary labels (§5–6). (3) A detection-limit analysis: a semi-synthetic positive control with a well-specified reference learner shows how strong a content preference must be for these methods to find it and that the limit lies in finding the content dimension; on MultiPref, annotators' own aspect labels show that value conflicts — where individual trade-offs could change a ranking — occur in 4 % of comparisons (§5.6–5.7). Supporting measurements of where the predictable structure sits are in §3 and Appendix D; protocols, amendments and failed hypotheses in Appendices A, B and E.

2. Related work

Pluralistic and distributional reward models. Distributional preference learning (Siththaranjan et al., 2024), diverging-preference detection (Zhang et al., 2025), PRISM (Kirk et al., 2024), MultiPref (Miranda et al., 2025) and DICES (Aroyo et al., 2023) motivate modelling disagreement; jury learning (Gordon et al., 2022) and multi-annotator models (Mostafazadeh Davani et al., 2022) predict individual annotators, as do disagreement deconvolution (Gordon et al., 2021), per-rater toxicity models (Kumar et al., 2021), demographic and opinion-based prediction of unseen annotators (Fleisig et al., 2023) and multilevel models of DICES raters (Homan et al., 2024). Graded preference strength enters reward training as a margin (Touvron et al., 2023; Wang et al., 2025), and ties have classical paired-comparison models (Rao & Kupper, 1967; Davidson, 1970). **Personalised reward models** infer a user latent from a few examples (Poddar et al., 2024; Chen et al., 2025), represent users in a low-rank reward subspace (Bose et al., 2025), induce persona prompts from a user's history (Ryan et al., 2025, who also release a benchmark of PRISM and Chatbot Arena users), learn per-user embeddings jointly with the reward model (Li et al., 2024), fit a new user as a combination of shared base rewards (Shenfeld et al., 2025; Luo et al., 2025), or infer the user with neural processes (Kobalczyk et al., 2024); related work personalises the policy (Jang et al., 2023; Feng et al., 2024), adapts to groups (Zhao et al., 2024; Chakraborty et al., 2024) or studies personalisation versus aggregation (Park et al., 2024). Value profiles condition a rating decoder on a rater's demonstrations and are tested on held-out raters (Sorensen et al., 2025). Benchmarks often simulate users by personas or mixtures of reward models (Castricato et al., 2025; Zollo et al., 2025); real multi-rater data with deliberately diverse candidates are emerging (Zhang et al., 2026). VPL's latent can collapse under sparse preference data (Kim & Kim, 2026). A multi-faceted comparison of personalisation methods (Dong et al., 2025) reports large differences between methods and safety costs. Annotator embeddings (Deng et al., 2023) and per-annotator heads (Mostafazadeh Davani et al., 2022) are close relatives of our offsets; our contribution is not the offset model but the new-rater test and what it shows. Rater groups have been found to use safety severity scales differently (Mishra et al., 2026), which is consistent with the leniency component we find on DICES. **Annotator demographics** as predictors of labels have been found weak in several studies (e.g. Orlikowski et al., 2023). **Measurement theory**: rater severity and scale use are standard facets in many-facet Rasch measurement (Linacre, 1989; Eckes, 2011) and response styles are a classical survey-methods topic (Baumgartner & Steenkamp, 2001). We connect the last line to the first: for new-rater prediction, the measurement-theory rater facet is what the reward model is missing. **Collaborative filtering and annotator models.** User and item bias terms carry much of the predictive gain in collaborative filtering (Koren et al., 2009), and user-specific ordinal thresholds predict personal rating distributions (Koren & Sill, 2011); annotator models estimate rater-specific labelling behaviour (Dawid & Skene, 1979; Raykar et al., 2010; Hovy et al., 2013), and the label-variation literature argues for modelling it (Plank, 2022; Uma et al., 2021). Our offsets are of this kind and are not new; what is new is testing personalised reward models against them on new raters, and measuring what content personalisation these models could detect at all.

3. Data, and where the predictable structure sits

Data. MultiPref (Miranda et al., 2025; 10,461 comparisons; 4 ratings each by 2 crowd and 2 expert raters; 227 persistent raters; 5 ordered levels A-clearly … B-clearly). PRISM (Kirk et al., 2024; 68,371 responses each rated 1–100 by one of 1,396 users; binned to 5 fixed levels). DICES-350 (Aroyo et al., 2023; 350 conversations fully crossed with 104 quality-checked raters; safe / unsure / unsafe).

A learner-free measurement (Appendix D). Before fitting any personalised model we measured, on MultiPref, where held-out predictable structure sits: beyond item location, rater-specific thresholds improve held-out likelihood about 25 times more than item-specific dispersion (+0.036 vs +0.0014 nats per rating), the item-level shape of the rat-

ing distribution that replicates across disjoint rater panels is small (split-half reliability 0.25), and about a tenth of it is decodable from text. Beyond item location, the predictable structure is therefore mostly rater-side scale use, while the item-level shape that replicates across disjoint panels is small and only partly in the text. These two facts come from different estimands and do not say that disagreement is "not in the text". The stage-1 decision rule we had written down returned **"mixed"**, because G/X was not ≤ 10 % (interval 5.7–11.9 %); the factorised model of §4 was chosen with that outcome known.

4. A three-parameter scale-use baseline

(Figure 2 illustrates the effect of the offsets.) Let $\ell(x) \in R^K$ be the class logits of a text-only distributional reward model for item x (K ordered rating levels). For rater r we add a text-independent offset vector: $p(y \mid x, r) = \text{softmax}(\ell(x) + b_r)$. Variants: **DF** ($b_r \in R^K$, free), **DFs** (two-sided preference scales only: b_r tied symmetrically, (e, s, t, s, e) — extreme, slight, tie — 3 parameters, of which 2 are identifiable because the softmax is invariant to a common shift; no A-vs-B position term). Rater offsets have a Gaussian prior centred on the training-rater mean; a new rater's offsets are the MAP estimate from k of their ratings with the text model frozen. Baselines under the same protocol: **D** (text-only distributional RM), **DG** (group offsets: crowd/expert, race, region), **DL** (low-rank text-interacting latent: $\ell(x) + (Wx + w_0) \cdot z_r$, d = 8, the personalised-RM family). An interpretable probit family (rater shift / log-scale on an ordered-probit text head) is reported as a secondary analysis.

Training and adaptation. The text model and the training raters' offsets are fitted jointly by penalised maximum likelihood,

$$\min \sum_{(i,r,y)} -\log p(y \mid x_i, r) + \lambda_{\text{text}} \|V\|^2 + \lambda_r \sum_r \|b_r\|^2,$$

with λ_{text} chosen on the text-only model and λ_r on held-out items of the training raters. A new rater r^* is adapted from k ratings $\{(x_j, y_j)\}$ by $\hat{b}_{r^*} = \arg\min_b \sum_j -\log \text{softmax}(\ell(x_j) + b)_{y_j} + \lambda_r \|b - \bar{b}\|^2$, with $\bar{b}$ the training-rater mean (the prior mean; $\hat{b} = \bar{b}$ at $k = 0$) and the text model frozen — a convex problem in 3–5 parameters, solved in milliseconds. Any text model that outputs graded log-probabilities can be used: we use a ridge head on frozen LLM features (§5.1–5.2), a LoRA-fine-tuned LLM reward model (§5.3) and the decoder of a published personalised reward model (§5.4).

5. Experiments

Protocol. 5 folds; in each, 20 % of raters (stratified by group) and 20 % of prompt groups are held out; models are trained on training raters $\times$ training items with penalties chosen on a validation split; each test rater is adapted on $k \in \{0, 5, 10, 20, 50\}$ of their ratings of *training* items (5 random draws) and evaluated on their ratings of *held-out* items. Primary metric: held-out log-likelihood per rating; also RPS and preference-direction accuracy; rater-cluster bootstrap (B = 2,000). Because test raters share test items on MultiPref and DICES, the headline k = 20 contrasts were also re-estimated with a two-way bootstrap that resamples raters and prompt groups independently: on graded labels every log-likelihood interval widens by at most 0.007 on either side and no log-likelihood conclusion changes; direction-accuracy differences of about a point or less (the style model's +0.5 on MultiPref, VPL's direction advantage over the style model) lose significance and are marked where they appear. Text model in §5.1–5.2: frozen Qwen2.5-7B features with a ridge head; §5.3: a LoRA-fine-tuned reward model.

5.1. Main result (MultiPref)

Figure 1 shows the gain over the text-only reward model as a function of k on all three datasets. In the table, DF/DFs/DG use the training-rater prior and DL and the DFs − DL column use the few-shot-tuned prior (the DFs − DL contrast is post hoc).

Preference-direction accuracy (D: 0.674): DFs +0.005 [+0.001, +0.008] at k = 20 (not significant once items are also resampled: [−0.001, +0.011]); DF −0.014 [−0.020, −0.007]. A three-number style model gains as much as a tuned 8-dimensional text-interacting latent (tie at k = 20) and more at k = 5, 10 and 50; fitting a rater's A-vs-B position from few ratings costs direction accuracy.

Reference points. To size these gains: on MultiPref at k = 20 a uniform forecast scores −1.609 nats per rating, a forecast that ignores the text and uses only the new rater's 20 ratings −1.487, the text-only model −1.441, the text model with the three offsets −1.342, and offsets fitted in-sample on all of the rater's held-out ratings (an optimistic ceiling for text-independent offsets) −1.262. Twenty ratings of a person, without the text, recover 73 % of the text model's gain over uniform; the offsets close 56 % of the gap between the text-only model and the in-sample ceiling. On DICES the rater's ratings alone predict better than the text alone (−0.766 vs −0.840 at k = 20). Even the in-sample ceiling raises MultiPref direction accuracy by only 1.3 points (0.674 → 0.687), confirming that text-independent offsets cannot carry ranking.

5.2. Replications (k = 20)

DF − D and DG − D use the training-rater prior, DF − DL the few-shot-tuned prior. Direction accuracy on PRISM means good-vs-bad around the scale midpoint.

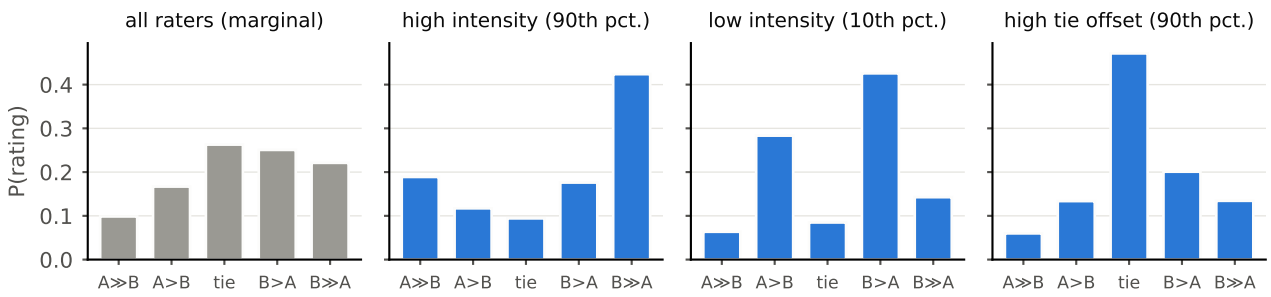


Figure 1. What three numbers per rater change. Left: the marginal distribution of MultiPref ratings. Right: the same distribution passed through the fitted scale-use offsets (extreme, slight, tie) of three real new raters (k = 50) closest to the 90th / 10th percentile of intensity (extreme minus slight) among raters whose tie offset is at or below the median, and closest to the 90th percentile of the tie offset. The offsets are text-independent: they reshape how strongly a rater is predicted to prefer, not which side.

k	DF − D	DFs − D	DG − D	DL − D (tuned prior)	DFs − DL (tuned, post hoc)
5	+0.028 [+0.012, +0.044]	+0.048 [+0.029, +0.070]	≈ 0	+0.023 [−0.002, +0.047]	+0.026 [+0.016, +0.037]
20	+0.080 [+0.055, +0.106]	**+0.100 [+0.074, +0.128]**	≈ 0	+0.095 [+0.069, +0.122]	+0.0045 [−0.0004, +0.0092]
50	+0.114 [+0.087, +0.142]	+0.128 [+0.101, +0.156]	≈ 0	+0.120	+0.007 [+0.002, +0.013]

5.3. On a fine-tuned reward model

With a LoRA-fine-tuned Qwen2.5-7B reward model (D-LoRA, trained per fold on the training raters' rating distributions) as the text model, the three-number style model still adds +0.100 [+0.071, +0.130] nats per rating at k = 20 — the same as its gain on the frozen-feature head (+0.100). Fine-tuning improves the text-only model by +0.027 [+0.016, +0.039] (direction accuracy +1.4 points) and the style model by the same +0.027: the two gains add. Personalising the frozen-feature model with 20 ratings per rater gains more than fine-tuning the text model (+0.072 [+0.041, +0.106]). DFs keeps the direction accuracy of D-LoRA (+0.002 [−0.002, +0.006]). Against in-context personalisation — a separate LoRA model on the same base, trained with ten of the rater's other ratings in the prompt and evaluated with the new rater's k ratings in the prompt — the three offsets on D-LoRA are better at k = 20. The in-context model's result varies with its training seed (two of fifteen runs failed, gaining −0.078 and −0.055 over D-LoRA against +0.021 to +0.125 for the others; the two failed runs have the highest training loss), so it was trained with three seeds per fold, and we fixed in advance that "better" requires a lower bound above zero for every seed and for the seed average. Averaged over seeds — the expected result of one training run — the offsets are better by +0.041 [+0.027, +0.054] nats per rating (209 raters; per seed +0.062, +0.018 and +0.042, every lower bound at least +0.005) and more accurate on which response the rater prefers (+1.9 points [+0.9, +2.7]). The in-context model does learn from the history (+0.051 [+0.024, +0.079] over D-LoRA on average), at several thousand prompt tokens per prediction, but its direction accuracy is below D-LoRA's (−1.7 points [−2.6, −0.8] on average; −0.6 to −3.0 per seed).

5.4. Published personalised reward models

We reimplemented VPL (Poddar et al., 2024) and PAL-B (Chen et al., 2025) from their official code as heads on the same frozen per-response embeddings and evaluated them under the same new-rater protocol, refitting the text-only and style models on those embeddings. On graded labels, at k = 20 the style model is better than VPL by +0.092 [+0.068, +0.119] nats per rating ([+0.063, +0.124] when prompt groups are also resampled) and than PAL-B by +0.158. VPL's latent does carry rater information once its posterior collapse is prevented (KL weight selected as 0 or 0.001; +0.087 over its own unpersonalised version), but only lifts it to the level of the text-only model; PAL-B's few-shot user weights give no gain. Their user representations are unrelated to response style (cross-validated $R^2 \approx 0$ against the style offsets). On binary labels — the setting in which both methods were proposed — every method's personalisation gain is at most 0.002 nats (small but nonzero for PAL-B with K = 4, and for VPL and the style model at k = 50), and VPL's text head is the best binary model (the style model is 0.010 nats behind it and 0.005 behind the text-only model, differences of the size of item-sampling noise): the individual differences that are predictable for new raters live in the graded levels and ties that binary preference data discard. VPL's per-response reward head is slightly better at predicting *which* response is preferred (0.677 vs 0.668), an effect present without personalisation and not significant once prompt groups are also resampled (style model − VPL:

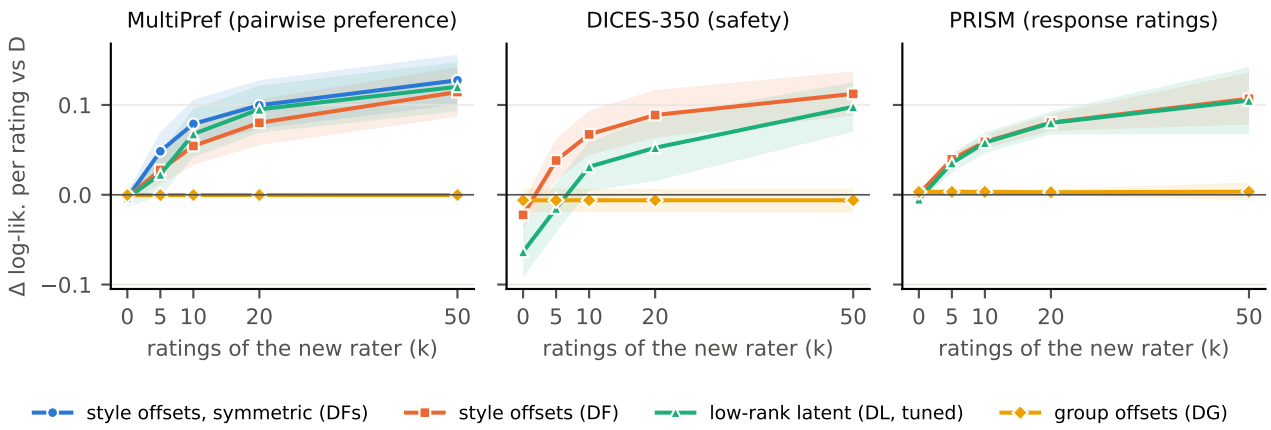


Figure 2. New raters: gain in held-out log-likelihood per rating over the text-only distributional reward model (D), as a function of the number k of the new rater's ratings used for adaptation; 95% rater-cluster bootstrap bands, 5 folds. Text-independent rater offsets (DF; symmetric three-parameter DFs on two-sided preferences) gain 0.08–0.10 nats per rating at k = 20 on all three datasets; group offsets (crowd/expert, race, region) gain nothing; a low-rank text-interacting rater latent with a few-shot-tuned prior ties the three-parameter DFs and beats the five-offset DF on MultiPref (DF − DL = −0.015). DF, DFs and DG use the training-rater prior; DL the few-shot-tuned prior.

dataset	DF − D	DG − D (groups)	DF − DL (tuned prior)
DICES-350 (safety)	+0.089 [+0.064, +0.117]	−0.006 (race)	+0.037 [+0.022, +0.053]
PRISM (ratings)	+0.080 [+0.071, +0.090]	+0.003 (region)	−0.001 [−0.006, +0.003] (formally not non-inferior: bound −0.006 < −0.005)

−0.9 points [−2.7, +0.8]). The offsets are also a plug-in: added to VPL (fitted on the same k ratings, VPL frozen) they improve it by +0.088 [+0.066, +0.111] nats per rating at k = 20, tie with the style model on likelihood (−0.004 [−0.014, +0.005]) and keep VPL's direction accuracy (+1.1 points over the style model alone; rater bootstrap only).

5.5. Summary of baselines (MultiPref, new raters, k = 20)

Gains are relative to each block's text-only model (D, Dp, D-LoRA); the LoRA block uses adaptation draw 0 only (the draw the in-context model was evaluated on), hence the style model's +0.092 here against +0.100 averaged over five draws in §5.3.

Rows within a block share the text model and the raters; across blocks the text models differ (text-only direction accuracy 0.674 / 0.670 / 0.688). VPL's and PAL-B's own gains over their unpersonalised versions are +0.087 and −0.000.

5.6. Could these methods have detected content personalisation? A semi-synthetic positive control

A ranking null is informative only if the methods would detect content-dependent preferences where they exist. We keep MultiPref's structure (raters, items, who rated what, folds and k-shot draws) and replace the labels by semi-synthetic ones. A generator — the text-only head plus scale-use offsets, fitted on all ratings — gives each rating a distribution, to which a rater-specific content term $(\theta_r^\top g_i)\,(-2,-1,0,1,2)$ is added to the logits; g_i are two features of the response pair (the log length ratio and a random direction in the per-response embedding space, orthogonal to the consensus quality direction) and $\theta_r \sim N(0, \tau^2 I)$. The scale τ is calibrated on the generator alone so that an oracle that knows θ_r gains +1.7, +5.2, +10.1 or +21.1 points of direction accuracy on the test ratings; $\tau = 0$ is a null control. Every method is then run under the paper's protocol. Because a method whose unpersonalised state is not the population state can improve with anyone's ratings, the contrast is **the method adapted on the rater's own k ratings minus the same method given another test rater's adapted parameters** (a random derangement within fold, k and draw). Before the runs we fixed that the protocol has power at a strength if some content-interacting method has a lower 95 % bound above +1 point at k = 20; the +10 and +21 panels and the untuned-prior low-rank runs were added after the first results and are exploratory.

Own minus another rater's k = 20 ratings, direction accuracy in points, 95 % rater bootstrap; 209 new raters, 5 folds. The low-rank latent uses the few-shot-tuned prior. k = 50 contrasts use the 180 of 209 raters with at least 50 training-

text model	personalisation	gain in LL per rating vs the block's text-only
frozen features (joint item embedding)	group offsets (crowd / expert)	≈ 0
	low-rank latent, tuned prior	+0.095 [+0.069, +0.122]
	style offsets, 3 numbers (DFs)	**+0.100 [+0.074, +0.128]**
frozen features (per-response embeddings)	VPL (reimplemented)	+0.002 [−0.008, +0.012]
	PAL-B (reimplemented)	−0.065 (worse)
	DFs	**+0.094 [+0.069, +0.120]**
	VPL + style offsets	+0.090
LoRA-fine-tuned Qwen2.5-7B (209 raters)	in-context rater history (RC-LoRA, mean of 3 training seeds)	+0.051 [+0.024, +0.079]
	DFs	**+0.092 [+0.063, +0.125]**

panel	oracle gain	low-rank latent (DL)	VPL	PAL-B	PAL-B, full features
null control ($\tau = 0$)	0	−0.10 [−0.38, +0.17]	−0.04 [−0.08, +0.01]	−0.23 [−0.45, +0.00]	−0.04 [−0.25, +0.17]
injected, $\tau 2$	+1.7	−0.01 [−0.29, +0.26]	+0.01 [−0.07, +0.10]	−0.10 [−0.36, +0.15]	—
injected, $\tau 5$	+5.2	−0.02 [−0.28, +0.26]	+0.01 [+0.00, +0.03]	+0.23 [−0.04, +0.51]	+0.26 [−0.14, +0.66]
injected, $\tau 10$ (exploratory)	+10.1	+0.09 [−0.18, +0.36]	+0.27 [+0.02, +0.54]	+0.42 [−0.08, +0.96]	+1.95 [+1.16, +2.74]
injected, $\tau 20$ (exploratory)	+21.1	+1.47 [+0.83, +2.18]	+6.92 [+5.63, +8.30]	+4.62 [+3.47, +5.78]	+8.13 [+6.54, +9.77]
real MultiPref	—	+0.30 [−0.00, +0.60]	−0.01 [−0.04, +0.02]	−0.43 [−0.72, −0.15]	+0.11 [−0.16, +0.37]

item ratings. PAL-B's own − other contrast is negative with compressed features (null control −0.23, real data −0.43): its per-rater adaptation adds noise there; with full features it is at zero (−0.04, +0.11). The injection uses one draw of θ and of the labels, so seed variability of the synthetic panels is not in the intervals. Last column: PAL-B on the full 3,584-dimensional features instead of their PCA-256 (exploratory; $\tau 2$ not run).

No method has power at the pre-specified strengths. In the exploratory dose-response, VPL and PAL-B detect content preferences worth +21 points but recover less than half a point of +10; the low-rank latent, whose user term interacts with a joint embedding of the comparison, has a signal at +21 points that stays below the threshold, and with the untuned prior it shows nothing up to +5 points either. On the real data VPL and PAL-B are at the null-control level, and the low-rank latent's marginal +0.30 [−0.00, +0.60] is matched by the text-independent offsets (+0.32 [+0.06, +0.58] on the same contrast), so it reflects scale use (which can tip a few near-tie predictions), not content. Read together: on MultiPref, content-level heterogeneity strong enough for an oracle to gain about 20 points of direction accuracy is excluded; heterogeneity worth up to about 10 points cannot be seen by these methods from 20 ratings, so the ranking null of §6 says nothing about it. Because the compressed features might be what limits the content-interacting methods, we re-ran VPL and PAL-B on the full features (exploratory). The text-only fits improve (k = 0 log-likelihood: PAL-B −1.533 vs −1.558, VPL −1.495 vs −1.574), but VPL's latent then collapses and is unused at every strength (own − other exactly 0), so this check rests on PAL-B. Its detection limit drops: it passes the threshold at +10 points; at +5 points its k = 50 gain is positive (+1.05 [+0.37, +1.72]; real data +0.19 [−0.12, +0.48]) but meets the +1-point criterion in only one of four synthetic draws. The real data stay at the null level (+0.11 [−0.16, +0.37] at k = 20). With this stronger detector, content heterogeneity of the injected form worth about 10 oracle points of direction accuracy is excluded at k = 20; at k = 50 the real data lie below the +5-point panel but the intervals overlap, so about 5 points is not excluded; the compressed features used for the main tables make the published methods weaker detectors than they need to be. **How far are these methods from what the data allow?** A well-specified reference learner — the injection's two features known, a per-rater weight vector with an empirical-Bayes prior, style offsets re-fitted, the same permutation contrast — recovers the injected preference from 20 ratings at every strength. On the oracle's scale (the rater's own weights against the population state, $\theta = 0$ here, with the rater's style offsets kept) it gains +0.89 [+0.40, +1.37] points at +1.7 oracle points, +3.1 [+2.3, +4.0] at +5.2 and +8.0 at +10.1, i.e. 52–79 % of the oracle; own − other roughly doubles these numbers, because another rater's weights are anti-informative, and is therefore not used for this learner. The statistical detection limit is therefore between +1.7 and +5.2 points, an order of magnitude below that of the published methods: what they lack is not ratings but the content dimension along which a rater differs. Results are stable over four synthetic draws, and the synthetic panels are noisier than the real data, so the calibrations transfer only approximately (Appendix B).

5.7. How much room is there? Value conflicts are rare

The positive control asks what the methods could detect; MultiPref also lets us ask how much there is to detect. Every MultiPref annotation carries the annotator's own helpfulness, truthfulness and harmlessness preferences next to the overall one, so a rater's way of trading these values off is observable. We fit an ordinal model of the overall preference from the annotator's own aspect labels (their values and

absolute values) with rater-specific cut-points (scale use) and either a pooled trade-off or a rater-specific trade-off, adapted on k of the rater's training-item annotations and evaluated on held-out items **with the rater's own aspect labels as inputs** — an oracle for the content side. The pooled trade-off weights harmlessness most (6.1, five-fold mean), then helpfulness (4.0) and truthfulness (2.0). With the rater's own aspect labels as inputs the pooled model already gets 97.5 % of directions right (74 % on conflict items): aspect and overall labels are produced together by the same annotator, so this ceiling concerns a small residual and is a property of MultiPref's concurrent labelling.

Even with the rater's own aspect judgements and all of their training ratings, a rater-specific trade-off predicts which response they prefer only 0.1 points better than the population's trade-off, and even fitted in-sample only 0.4 points better. The reason is that the aspects point in opposite directions in only 4.0 % of the held-out non-tie comparisons (3.5 % of all annotations); there individual trade-offs matter a great deal (in-sample +9.7 points; among the 20 raters with at least ten helpfulness–harmlessness conflicts, the share of conflicts resolved in favour of helpfulness ranges from 0.00 to 0.94), but they are too rare to move overall accuracy. Individual trade-offs do change how strongly raters prefer (up to +0.046 nats with all their ratings), in line with the calibration-not-ranking pattern. We fixed in advance that an aspect-weight personalised reward model would be built only if the lower bound of this ceiling exceeded half a point with all training ratings; it does not, so we did not build one. The learned personalisers cannot be tested on the conflict items either: the 243 held-out non-tie ratings on them give own − other intervals wider than the achievable ceiling (k = 20: low-rank latent +0.1 [−2.2, +2.4]). **A calibrated test on the real data.** The ceiling above gives each rater's own aspect labels; a deployable personaliser would know only the item's aspects. We therefore ran the reference learner of §5.6 on the real labels with the item's consensus aspect differences from its *other* annotators as content features (3-d), with a hierarchical prior centred at the population weighting (estimated on the training raters) and compared each rater's own weights with that population weighting (style offsets kept). Individual weights transfer *worse* than the population's: −0.28 [−0.56, −0.06] points at k = 20 and −0.63 [−1.01, −0.25] at k = 50. With injected aspect-weight heterogeneity it gives +0.46 [−0.00, +0.96] at k = 50 when an oracle would gain +1.4 points and +3.4 [+2.6, +4.3] at +5.0. The real data lie well below the +5-point panel and, at k = 50, just below the +1.4-point panel: transferable heterogeneity in how raters weigh these three values is at most of the order of one to two oracle points on MultiPref. This test is exploratory and post hoc, and its contrast was revised four times (Appendix A; no version shows a positive transferable aspect part), the calibration strengths were chosen after seeing the real bound, it uses one synthetic draw on panels noisier than the real data, and it uses other annotators' aspect labels of the test items (an aspect oracle, not a deployable model). Three caveats: the ceiling bounds personalisation of the *weighting* only — disagreement about which response is more helpful or harmless is absorbed into the inputs; it covers content preferences expressible through these three aspects only; MultiPref's prompt sampling determines how often values conflict; and annotators may have aligned their aspect labels with their overall label, which would understate conflicts.

6. What personalisation buys: calibration, not ranking

Binary labels remove the signal. On MultiPref with ties dropped and labels binarised, every method's personalisation gain is at most 0.002 nats per rating (§5.4), and the scale-use baseline's binary accuracy equals the text-only model's (−0.0002). The individual differences that are predictable for new raters live in the graded levels and in ties — exactly what binary preference data discard, and the format in which most personalised reward models are trained and evaluated. **Ranking moves little.** Text-independent offsets change how strongly a rater is predicted to prefer a response, not which response: on MultiPref the scale-use baseline raises direction accuracy by only +0.005 [+0.001, +0.008] at k = 20 (about half of it already at k = 0), while RPS drops from 0.1593 to 0.1532. On single-response safety ratings the same offsets do move the safe/unsafe decision (+4.8 points on DICES), because there a rater's threshold *is* the decision. Content-interacting personalisation does not do better on ranking: VPL's advantage in direction accuracy is present without personalisation (k = 0), comes from its text head, and is within item-sampling noise. Text-only direction accuracy on MultiPref (0.674) is already as high as the agreement in direction between two human raters of the same item (0.673 [0.666, 0.680]): a population model predicts a new rater about as well as another person does. Against its own k = 0 version, no method's direction accuracy on MultiPref rises by more than 0.2 points at k = 20, and every upper 95 % bound stays below +0.8 points even when prompt groups are resampled together with raters (VPL +0.07 [−0.57, +0.74], style model −0.10 [−0.69, +0.54], PAL-B −0.48 [−1.11, +0.14]). **The ranking null replicates on real users' own choices.** PRISM users each saw 2–4 responses to their own prompt, scored them and chose one; pairing the chosen response with every other one gives 41,199 native pairwise choices of 1,396 users. Under the same new-user protocol (5 user folds × 5 conversation folds, k of the user's pairs from training conversations), we fixed before any result that the ranking null would be stated only if every method's improvement in choice accuracy over its own k = 0 version had an upper 95 % bound below one

rater ratings used	rater − pooled trade-off, direction accuracy (points)	log-likelihood per rating	on conflict items (points)
k = 20	+0.03 [−0.05, +0.12]	+0.011 [+0.003, +0.020]	+0.6 [−1.1, +2.4]
k = 50	+0.06 [−0.09, +0.21]	+0.023 [+0.011, +0.037]	+0.3 [−2.7, +3.8]
all training items	+0.10 [−0.24, +0.43]	+0.046 [+0.028, +0.068]	+2.0 [−5.0, +9.2]
in-sample (fitted on the test items)	+0.44 [+0.27, +0.64]	+0.094 [+0.079, +0.112]	+9.7 [+5.6, +14.5]

point at k = 20. It holds for all of them (1,033 new users at k = 0; the 669 with at least 20 adaptation pairs at k = 20): VPL +0.12 points [−0.08, +0.33], PAL-B +0.0 [−0.4, +0.4] (K = 2) and +0.3 [−0.1, +0.7] (K = 4), the 5-offset model −0.6 [−1.4, +0.2]; the symmetric scale-use model is identical to the text-only model on binary choices, because a binary label carries no intensity. LoRe (Bose et al., 2025), which reports +10 points for unseen PRISM users, run here with its own few-shot fit and with a validated adaptation ridge under the same pre-set rule, also fails it: own gain at k = 20 −1.5 [−2.5, −0.4] (regularised −0.5 [−1.2, +0.2]), own − other +0.2 [−0.8, +1.2]. In LoRe's own setting (its embeddings, split and pair construction) we obtain 67.4 % for the monolithic model and 63.6 % for LoRe (published 61.0 % and 71.0 %; Appendix B): the published gain is not reproduced. With a graded label cut from the users' own score differences (thresholds fixed in advance), the MultiPref pattern returns at half the size: at k = 20 the scale-use model gains +0.043 [+0.031, +0.057] nats per rating over its k = 0 version, VPL +0.015 [+0.005, +0.027], PAL-B 0.000; the scale-use model is ahead of VPL by +0.045 [+0.030, +0.062]; and no method's own direction-accuracy gain exceeds +0.2 points (the 5-offset model loses 1.7).

7. Recommendations

(i) Collect and keep graded labels (and ties) when individual differences matter; binary preferences erase most of what is predictable about a new rater. Sample comparisons for value conflicts (for example, pairs on which aspect ratings disagree): in MultiPref only 4 % of comparisons involve one, and they are where individual trade-offs matter. (ii) Report a scale-use calibration baseline (a few text-independent offsets per rater) next to any personalised reward model. Beating it on likelihood is necessary but not sufficient evidence of content personalisation; a method that learns content preferences should show it in ranking or choice outcomes, where the offsets cannot help by construction. (iii) Evaluate on raters unseen in training, adapted from k of their own ratings, and on held-out items, and compare each method with itself given another rater's ratings (a permutation control): a poor unpersonalised state inflates gains measured against k = 0. (iv) Report calibration (log-likelihood, RPS) and ranking (direction accuracy) separately: they answer different questions — how strongly someone prefers, and what to show them. (v) Check what a learned user representation encodes; in the two published models we reimplemented it was unrelated to scale use ($R^2 \approx 0$) and carried little else that transferred to new raters.

8. Discussion

Why did published personalised models learn little about new raters here? Their evaluations differ from ours in three ways that each favour content-dependent personalisation: synthetic users whose preferences differ in content by construction, binary labels that remove scale use, and users seen during training. On real raters, graded labels and unseen users, the structure that transfers from k ratings to new items is mostly how the rater uses the scale, which neither VPL's latent nor PAL-B's prototypes represent ($R^2 \approx 0$). Adding the offsets to VPL recovers the whole gain (+0.088): the two are complementary, and a content-personalising method should be shown to beat VPL plus offsets.

9. Limitations and analysis log

Protocols, amendments and failed hypotheses: Appendices A, B, E. The main tables use frozen-feature text heads (a LoRA reward model only in §5.3), and compressed features make the published methods weaker detectors than full features. Our VPL, PAL and LoRe are reimplementations as heads on frozen embeddings; VPL's and PAL's published numbers were not reproduced in their native settings, and LoRe's PRISM gain is not reproduced even in its own setting with its embeddings, for reasons we could not isolate. The positive control injects low-dimensional preferences on MultiPref only, and its panels are noisier than the real data, so detection limits transfer only approximately. The aspect analyses rest on MultiPref's concurrent aspect labels. All datasets are English, and the per-rater calibration is not shown to change a downstream decision.

Impact Statement

This paper evaluates personalised reward models, which are meant to let language models serve people whose preferences differ. Its main finding — that what such models learn about a new rater from few ratings is mostly how that rater uses the rating scale — bears on claims that current methods represent individual or minority values: reported personalisation gains can reflect scale use rather than con-

tent preferences, and evaluations on synthetic users, binary labels or seen users can overstate what is learned. We see no direct negative use of the protocol. The datasets are public, were collected with consent by their creators, and are used under their licences; we use rater identifiers only as anonymous indices and report no individual-level results.

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A. Protocols and amendments

All analyses were specified in timestamped internal protocols (PROTOCOL_STAGE1, PROTOCOL_STAGE2, PROTOCOL_STAGE2B, PROTOCOL_STAGE2C, PROTOCOL_STAGE3); stages 2b and 2c were written after the stage-2a results. Changes after a result were logged as numbered amendments with the reason; the main ones: the first decoding split (a = b = 2) was structurally empty (two ratings determine their distribution) and was replaced by a = 3, b = 1; hand-set penalties in the rater–item model over-fitted and were replaced by inner cross-validation (which reversed an earlier, wrong conclusion that rater shifts do not help); penalty grids were extended when selections fell on grid edges, before any hypothesis was read; the low-rank baseline lacked a constant column and was fixed in its favour; the probit group model had a scale non-identifiability and was fixed; VPL suffered posterior collapse and its KL weight was re-selected including 0; PAL-B's adaptation ridge was re-selected by k-shot validation. Every superseded full run is kept and reported in the supplementary results; two debugging smoke runs and one fold whose output was deleted unread when the penalty grids were extended are not. The real-data aspect contrast of §5.7 was revised four times: own − other; with a style-only control after a first, confounded +0.45 [+0.01, +0.94]; against $\theta = 0$; and against the population weighting, after the reference learner turned out to shrink towards 0 rather than towards the population. The few-shot-tuned low-rank prior was selected at the top of its grid (8192) in most folds; extending the grid to 524,288 left every DL result unchanged in substance (real data: own − other +0.31 vs +0.30 at k = 20, DFs − DL +0.003 vs +0.0045; synthetic +5/+10-point panels: +0.10 / +0.09 vs −0.02 / +0.09).

B. Hypotheses and outcomes

Stage 2a (MultiPref, k = 20): H1 DF > D — holds (+0.080); H5 (probit F1 > D, the primary probit contrast after Amendment 1) — not supported (+0.020 [−0.001, +0.041]; homoscedastic F − D −0.029), superseded by H1 in Amendment 2; H2 DF > group — holds; H3 DF ≥ low-rank − 0.005 — holds under the training-rater prior, not robust under a few-shot-tuned prior (−0.015); H4 rater scale > rater shift — holds on MultiPref only; H6 symmetric style model gains likelihood without losing direction accuracy — holds and robust. Replications: H1r, H2r hold on DICES and PRISM; H3r holds on DICES (robust) and is a tie on PRISM under the tuned prior. Stage 2b: B1, B2, B3 hold. Stage 2c: C1 holds (graded), C2 holds (binary: |DFs − D| accuracy −0.0002; all personalisation gains ≤ 0.002 nats), C3 fails (published latents do not encode style, $R^2 \approx 0$), C4 not met as a conjunction (first part holds, +0.088 over VPL; second part a tie with DFs, lower bound −0.014); Stage 3 (positive control): $\tau 0$ control passes; power at the pre-specified strengths (+1.7, +5.2 oracle points) not established — no content-interacting method's own − other gain has a lower bound above +1 point; exploratory dose-response: VPL and PAL-B pass at +21 points, none at +10; the split-half content estimator (S3-C) has no power; with full features (exploratory, Amendment 5) the $\tau 0$ control passes and PAL-B passes at +10 points (k = 20), so the PCA bottleneck explains part of the detection limit. R1 (PRISM native pairwise choices: every method's own choice-accuracy gain at k = 20 has an upper bound < +0.01) holds (largest bound +0.0071, PAL-B K = 4).

Scale use and the shared reward model. Training the text model jointly with rater offsets and then using the population-mean offsets (k = 0) does not give a better consensus model than ordinary training: −0.003 [−0.010, +0.003] nats on MultiPref (direction +0.3 points, n.s.), −0.022 [−0.037, −0.008] on DICES, +0.001 on PRISM, −0.009 [−0.016, −0.004] with the LoRA reward model. Ordinary training averages scale use out; it matters for predicting a person, not for learning the content preference.

What the offsets encode. In the probit family, rater *scale* matters more than rater *shift* on MultiPref (+0.036 [+0.020, +0.057] at k = 20), but rater *shift* dominates on DICES (−0.053) and PRISM (−0.031): leniency/strictness on single-response rating scales, intensity on two-sided preference scales.

What this does and does not say about pluralism. Our results do not show that people agree about content: the item-level disagreement that replicates across rater panels is real (split-half reliability 0.25 on MultiPref) and about a tenth of it is decodable from text. They show that, for predicting an individual new rater from few ratings, that content-level disagreement is a small part of what is predictable, and that demographic or crowd/expert groups add nothing beyond the text.

Positive control: seed variability and realism (Stage 5). Over four synthetic draws (θ and labels) the reference learner is stable (+2.4 to +3.1 at +5 points, +6.4 to +8.0 at +10), PAL-B on full features detects the +10-point panels in all four (lower bounds +1.16 to +3.23) and the +5-point panels in none (+0.26 to +1.40), and the low-rank latent detects neither in any draw (−0.31 to +0.15). A limitation of all synthetic panels: they are noisier than the real data (rater-pair direction agreement 0.59 at $\tau = 0$ against 0.673 [0.666, 0.680]; Krippendorff's α 0.11 against 0.28), and adding item fixed effects to the generator did not close the gap (0.60), nor does it reproduce the small effect of style offsets on direction (+0.32 real vs −0.07 at $\tau = 0$); the calibrations transfer to the real data only approximately, with no claimed direction of bias.

LoRe on all panels (Stage 6). Across PRISM pairs and MultiPref, binary and graded labels and k $\in \{5, 20, 50\}$, LoRe never improves direction or choice accuracy over its own k = 0 version or over the text-only model (own gain -6.2 to +0.1 points; the regularised version -1.1 to -0.0), and its log-likelihood is below the text-only model's everywhere. Its own $-$ other is positive in some cells (MultiPref binary, k = 20: +1.3 [+0.5, +2.0]) while its own gain is negative there (-1.2 [-1.9, -0.5]): another rater's over-fitted weights are anti-informative, the scale problem discussed for the reference learner, so own $-$ other alone is not evidence of personalisation for methods whose few-shot fit over-fits.

Confirmatory and exploratory results. Confirmatory (rule fixed before the result): the stage-2a hypotheses H1–H6 and their replications; B1–B3; C1–C4; R1 (PRISM native choices); the Stage-3 positive control at +1.7 and +5.2 oracle points (no power); the Stage-4 ceiling rule (not met); the Stage-6 LoRe rule (PRISM null not withdrawn). Exploratory (added or changed after a result, labelled in the text): the symmetric offsets DFs and their hypothesis; the few-shot-tuned prior; the +10 and +21 panels and the full-feature runs; the reference learner and its oracle-scale contrast; the calibrated real-data aspect test; the regularised LoRe; the two-way bootstrap re-estimates.

Reproducing LoRe's PRISM result (Stage 6, Amendments 2–3). Unseen-user binary accuracy, mean $\pm$ sd over five random user splits, LoRe's native setting (users with at least six dialogues split 50/50, half of each user's conversations for few-shot and half for test, one chosen-vs-other pair per conversation; 1,267 users, about three few-shot pairs per unseen user):

Taking one pair per interaction instead (every turn; 20 pairs per user) gives 58.8 / 59.2 / 57.0 with our features. The remaining differences — a stronger monolithic baseline here and a few-shot fit that helps there but hurts here — could not be traced to any setting the LoRe paper reports. Under our protocol (user- and conversation-disjoint folds) the Skywork embeddings leave the conclusion unchanged: LoRe's own gain at k = 20 is -1.3 [-2.5, -0.2] points (regularised -0.5 [-1.3, +0.2]) and own $-$ other +0.2 [-0.9, +1.2], as with our features.

C. Implementation

Text features: last-token final-layer states of Qwen2.5-7B-Instruct under a 5-point judge prompt (MultiPref, PRISM) or a 3-level safety prompt (DICES), label-free PCA to 256 dimensions. Folds: 5 rater folds (stratified by group) $\times$ 5 prompt-group folds; validation: 15 % of training prompt groups. k-shot draws: 5 per k, identical across all models (RC-LoRA: draw 0 only). PRISM item folds are by conversation. Bootstrap: rater clusters, B = 2,000. LoRA: r = 16 on all attention and MLP projections, lr 10^{-4}, one epoch, accumulation 8. VPL / PAL-B: official architectures (latent 512, hidden 512; d_pref 512, mlp2-gelu projectors, angle learner) with an ordered-logistic link for graded labels; VPL's KL weight selected from $\{0.01$ (official), $0.001, 0\}$ and PAL-B's adaptation ridge by k-shot validation (non-official choices made to prevent collapse and over-fitting).

D. Where is the predictable structure? Estimators and results

Estimators. For an item with ratings on K ordered levels, let h be the empirical rating distribution and C the cumulative operator. Split the item's raters at random into disjoint panels P_1 (a raters) and P_2, and let q be a cross-fitted *location-only* forecast with exactly the mode and mean of P_1 (built from other items in the same mode-and-mean cell). With $r_1 = C(h_1 - q)$, $r_2 = C(h_2 - q)$, $M = \|r_1\|^2/n$ and $X = \langle r_1, r_2\rangle/n$, the forecasting value of the raw distribution over its own location obeys the identity $S(h_1) - S(q) = M - 2X$ under the ranked probability score, so $\rho = X/M$ is a split-half reliability of the distribution's *shape* and the raw distribution beats its location exactly when $\rho > 1/2$. Text decodability is the proper-score gain $G = (2\langle \hat{r}, r_2\rangle - \|\hat{r}\|^2)/n$ of a residual $\hat{r}$ decoded from frozen text features, cross-fitted over prompt groups, reported as the fraction G/X of the reproducible shape. The rater–item decomposition fits ordered-probit models $Y^*_{ir} = \theta_i + \delta_r + \sigma_i\sigma_r\varepsilon$ with item locations θ_i, rater shifts δ_r, rater scales σ_r and item dispersions σ_i, compares held-out likelihood over random cells (every held-out rater and item appears in training), and chooses all penalties by inner cross-validation.

Reproducibility. With the split-half estimator defined above, the reliability of the rating distribution's shape beyond its exact mode and mean is ρ = 0.25 [0.22, 0.29] on MultiPref overall preference with three raters (helpful 0.29, harmless 0.21): the raw few-rater distribution forecasts a new rater worse than its own location ($\rho < \frac{1}{2}$). **Text decodability.** A cross-fitted ridge decoder on frozen Qwen2.5-7B-Instruct features, scored on an independent rater panel, recovers G/X = 10.5 % [5.7 %, 11.9 %] of the reproducible shape (within-cell permutation control ≤ 0; positive control 0.35) — comparable to the upper end for ordinal quality ratings (upper bounds 0.4–10 %) and below NLI (17–26 %). **Rater vs item.** A long-format ordered-probit decomposition with real rater identities and penalties tuned by inner cross-validation: rater-specific thresholds improve held-out likelihood by +0.036 nats per rating over item location alone, item-specific dispersion by a further +0.0014 (ratio 25; 19–36 across the five folds); of the rater part, rater scale contributes +0.031 and rater shift +0.005.

embeddings	monolithic BT	LoRe, no adaptation	LoRe, few-shot fit
Skywork-Reward-Llama-3.1-8B-v0.2 (LoRe's)	67.4 ± 1.1	67.3 ± 1.8	63.6 ± 3.7
Qwen2.5-7B-Instruct (ours)	67.7 ± 1.5	68.7 ± 1.7	61.0 ± 2.5
published (Bose et al., 2025)	61.0 ± 1.2	—	71.0 ± 1.0

E. Protocols and what did not hold

All analyses are documented in an analysis log: timestamped internal protocols written before each stage (not public registrations, so they document rather than certify the order of decisions); changes after results are logged as amendments with reasons, and the symmetric style model DFs and its hypothesis were added after the first full result. Not supported or not robust: (i) the stage-1 decision rule returned "mixed" (text decodability not ≤ 10 %); (ii) non-inferiority of the 5-parameter DF to the low-rank latent (DF $-$ DL ≥ -0.005) holds with the training-rater prior but not with a few-shot-tuned prior on MultiPref (-0.015) and is formally not met on PRISM (bound -0.006); (iii) "scale beats shift" holds only on MultiPref; (iv) the originally registered probit-family hypothesis is not supported (F1 $-$ D = +0.020 [-0.001, +0.041], the primary probit contrast after Amendment 1; homoscedastic F $-$ D = -0.029 [-0.049, -0.008]; superseded when the primary family was moved to the distributional head); (v) the prediction that published user latents encode style failed ($R^2 \approx 0$); (vi) the hybrid's non-inferiority to DFs is formally not met (bound -0.014). Four baseline problems found during the study were fixed in favour of the baselines — a missing constant column in the low-rank latent, a scale non-identifiability in the probit group model, VPL's posterior collapse, and PAL-B's near-unregularised adaptation — and the runs before each fix are reported.